From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 2005–2024

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Abstract

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when Matosinhos ceased refining in May 2021. Using monthly JODI physical flows, the Eurostat oil balance, DGEG national statistics and the European Commission Weekly Oil Bulletin, this report measures how the product balance and price exposure changed. The 2013 upgrade produced a clear structural break: diesel refinery output rose and Portugal moved from a net diesel importer to a net diesel exporter, with the net-import-to-demand ratio breaking at 2013 ($F = 19.62$, Benjamini–Hochberg adjusted $p = 0.003$). After 2021 the direction reverses. Annual break tests at 2022 detect nothing, but they have only three post-event observations; a monthly design with month fixed effects separates the two episodes and finds both to be distinct. Diesel refinery output falls by 105 kt per month at the Matosinhos boundary ($p = 0.004$) and by a further 152 kt per month during the 2022 energy-stress period ($p < 0.001$), while the net-import-to-demand ratio rises by 0.20 and 0.38 respectively. Short-run pass-through from Spanish to Portuguese pre-tax diesel prices increases from 0.73 to 1.01 after the transition ($p < 0.001$). Gasoline short-run pass-through does not move ($p = 0.23$), but an error-correction model, which the cointegration diagnostic indicates for gasoline, shows its adjustment speed roughly tripling from -0.091 to -0.299 per week ($p = 0.004$), shortening the half-life of a price gap from about eight weeks to two. Both fuels therefore became more tightly linked to Spain, through different channels. 2022 confounds the post-closure window and this report does not claim a causal effect of the closure itself.

1 Research Question and Hypotheses

The empirical question is whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products and reduced supply resilience, and whether Portugal–Spain price co-movement changed after the closure.

Four mechanisms are separated rather than assumed to operate as one chain: refining capability, the production and trade balance, physical import dependence, and price co-movement. The pre-specified hypotheses are that the 2013 Sines hydrocracker changed diesel-producing capability and the trade balance; that the 2021 Matosinhos transition changed import dependence and domestic refinery-output coverage; that 2022 is a distinct supply-system stress episode rather than clean post-treatment evidence; and that any price change must be evaluated on pre-tax prices.

2 Data and Coverage

Four independent sources are used. Table 1 states what each covers over the 2005–2024 study window.

Table 1: Source coverage over the study window.

Source	Role	Coverage 2005–2024
JODI Oil World Database	Monthly physical flows: imports, exports, refinery output, demand	Complete. 160 of 160 annual product-flow cells, every year with 12 observed months
Eurostat `nrg_cb_oil`	Annual balance; Portugal and Spain	Complete for both countries, 160 of 160 cells each
DGEG trade workbooks	National import/export statistics	**2019–2024 only.** 2005–2018 is not published in this series
DGEG long sales workbook	Domestic market sales	Complete, 1970–2024
EC Weekly Oil Bulletin	Weekly pre-tax and post-tax consumer prices	Complete. 995 Portuguese and 994 Spanish weekly observations per product

2.1 The one material coverage gap

DGEG publishes annual product trade workbooks only from 2019. For 2005–2018 the trade cross-check therefore runs against Eurostat rather than against the DGEG primary publication. This is a weaker form of corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both are available. Eurostat is best understood as a second channel for the same national statistics, not as a third independent opinion. Domestic demand is unaffected: the DGEG long sales workbook covers the whole window, so demand has three sources throughout.

2.2 Product definitions

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `O4652`, motor gasoline. Aviation gasoline is excluded. In the DGEG sales workbook, gasoline is the *Gasolinas* total and diesel is road gasoil plus coloured and marked gasoil; road gasoil alone sits about ten per cent below the JODI and Eurostat demand concept. Differing biofuel treatment does not drive the residual differences between sources: the Eurostat blended-biofuel wedge is 0.0 kt on exports in every year of the window and at most 4.5% on imports.

3 Methods

For fuel j in year t, with imports M, exports X, domestic demand D and refinery output Q^{refinery}:

$$\text{NetImports}_{j,t} = M_{j,t} - X_{j,t}, \qquad \text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}},$$

$$\text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}}, \qquad \text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

The refinery-output-to-demand ratio is not self-sufficiency. Portuguese refinery output may be exported while domestic demand is met partly by imports, and for gasoline this is the normal state of the system.

Annual structural breaks use Chow tests on a linear trend at pre-specified years, with 2021 excluded as a transition year so it is assigned to neither side, and Benjamini–Hochberg adjustment across the sixteen tests. Monthly event timing uses a segmented regression on 293 monthly observations with month fixed effects, a calendar-month trend and HAC(3) standard errors. Each phase-trend interaction is centred on its phase boundary, so a phase coefficient is the level shift at that boundary against the extrapolated pre-closure trend, and its companion trend term is the change in slope within the phase. Price model family is selected from stationarity and cointegration diagnostics rather than assumed, with HAC(8) standard errors.

4 Physical Balance Evidence

4.1 Long-run description

Table 2 gives the annual diesel panel. Two regimes are visible. From 2005 to 2012 Portugal is a net diesel importer, with refinery output covering 73–92 per cent of demand. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative: Portugal manufactures more diesel than it consumes and exports the surplus. From 2021 the system returns to net import dependence, and by 2023 refinery output covers only 66 per cent of demand, the lowest value in the series.

Table 2: Portugal diesel annual balance, kt and ratios. Source: `fuel_annual_analytical_panel.csv`.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
2005	875	211	5,632	5,043	0.16	0.12	0.90
2006	635	314	5,471	5,055	0.12	0.06	0.92
2007	775	193	5,530	4,745	0.14	0.11	0.86
2008	1,006	164	5,450	4,671	0.18	0.15	0.86
2009	1,475	94	5,537	4,074	0.27	0.25	0.74
2010	1,167	35	5,601	4,273	0.21	0.20	0.76
2011	1,315	120	5,191	3,782	0.25	0.23	0.73
2012	892	349	4,678	4,059	0.19	0.12	0.87
2013	581	1,692	4,535	5,687	0.13	−0.24	1.25
2014	697	1,295	4,697	5,117	0.15	−0.13	1.09
2015	783	2,286	4,941	6,279	0.16	−0.30	1.27
2016	942	2,076	4,834	5,901	0.19	−0.23	1.22
2017	897	2,082	4,974	5,984	0.18	−0.24	1.20
2018	827	1,140	5,089	5,228	0.16	−0.06	1.03
2019	1,075	899	5,075	4,777	0.21	0.03	0.94
2020	875	1,304	4,403	4,606	0.20	−0.10	1.05
2021	1,491	1,378	4,728	4,199	0.32	0.02	0.89
2022	1,275	331	4,962	3,941	0.26	0.19	0.79
2023	1,597	285	5,084	3,348	0.31	0.26	0.66
2024	1,222	731	4,868	4,133	0.25	0.10	0.85

Gasoline behaves differently throughout. Portugal is a net gasoline exporter in every year of the window, with refinery output covering between 1.4 and 2.6 times domestic demand. This is the asymmetry the ratio definition warns about: a system can manufacture a large surplus of one light product while importing another.

4.2 The 2013 upgrade is a clear structural break

Table 3 reports the Chow tests. Three diesel outcomes break at 2013 and survive multiplicity adjustment comfortably. The break is in the direction the engineering implies: more diesel manufactured, more exported, and a net-import position that turns negative.

Table 3: Chow tests at pre-specified break years, 2021 excluded as a transition year. Source: `structural_break_tests.csv`.

Product	Outcome	Break	F	p	BH-adjusted p
diesel	net import / demand	2013	19.62	0.0002	0.0026
diesel	refinery output	2013	16.68	0.0003	0.0027
diesel	exports	2013	13.79	0.0008	0.0041
gasoline	refinery output	2013	5.23	0.0232	0.0867
diesel	imports	2013	4.95	0.0271	0.0867
No 2022 break test reaches significance; the smallest adjusted p is 0.378.					

The absence of a detected 2022 break should not be read as evidence that nothing happened. Each 2022 test has eight pre-event and three post-event annual observations, which gives very low power against anything short of an enormous effect. The monthly design below is the appropriate instrument for that period.

4.3 Monthly evidence separates the closure from the 2022 shock

Table 4 reports the segmented monthly model for diesel. Both phases carry independent, statistically distinguishable effects, which is the central reason a monthly design was required: annual data cannot separate a May 2021 event from a March 2022 one.

Table 4: Segmented monthly event model, diesel, $n = 293$ months, month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary. Source: `monthly_event_models.csv`.

Outcome	Term	Estimate	Std. error	p
Refinery output (kt/month)	Matosinhos transition	-105.21	36.02	0.004
	Energy stress 2022	-151.77	23.31	< 0.001
Imports (kt/month)	Matosinhos transition	42.23	23.40	0.071
	Energy stress 2022	28.36	11.22	0.012
Net import / demand	Matosinhos transition	0.2005	0.0713	0.005
	phase trend	0.0360	0.0153	0.019
	Energy stress 2022	0.3755	0.0554	< 0.001

Diesel refinery output falls by roughly 105 kt per month at the May 2021 boundary and by a further 152 kt per month in the 2022 stress period. The net-import-to-demand ratio rises by 0.20 at the closure and by 0.38 during the energy-stress phase, and the closure phase also carries a significant upward slope of 0.036 per month. Both episodes move the system in the same direction, and the 2022 effect is the larger of the two.

4.4 2022 as a stress episode

Against a 2013–2019 baseline, 2022 diesel exports of 331 kt sit 2.44 standard deviations below the mean and at the zeroth percentile of the baseline distribution, while diesel imports of 1,275 kt sit 2.74 standard deviations above and at the hundredth percentile. Gasoline exports in the same year are unremarkable ($z = -0.46$). The stress is diesel-specific, which is consistent with the documented suspension of Russian oil-product and vacuum gas oil purchases feeding Sines diesel manufacturing, and inconsistent with a general refining shutdown.

4.5 Descriptive change around the transition

Comparing the five years before 2021 with the three years after, and treating 2021 itself as a transition year, diesel gross import dependence rises from 0.190 to 0.274, a change of 8.4 percentage points. The diesel net-import-to-demand ratio rises from −0.119 to 0.183, a change of 30.2 percentage points, and refinery output coverage falls from 1.088 to 0.767. These are descriptive pre/post differences, not effect estimates.

5 Price Co-Movement

Model family was selected from diagnostics rather than assumed. For diesel, neither Portuguese nor Spanish pre-tax price levels reject a unit root ($p = 0.124$ and $p = 0.242$) and the pair is not cointegrated ($p = 0.240$), so a levels regression would be spurious and the short-run log-difference model carries the inference. For gasoline the levels are also non-stationary but the pair is cointegrated ($p = 0.000026$), so an error-correction model is indicated and estimated below. No ECM is fitted for diesel: with cointegration rejected, the disequilibrium term would not be a valid long-run residual.

Table 5: Short-run pass-through, $\Delta \log$ pre-tax price, HAC(8), n = 1,077 weeks. Sources: `price_short_run_models.csv`, `price_model_choice.csv`.

Product	Term	Estimate	Std. error	p
diesel	$\Delta \log$ ES	0.7348	0.0341	< 0.001
	$\Delta \log$ ES $\times$ post	0.2719	0.0541	< 0.001
gasoline	$\Delta \log$ ES	0.7997	0.0593	< 0.001
	$\Delta \log$ ES $\times$ post	0.0352	0.1375	0.798

Weekly pass-through from Spanish to Portuguese pre-tax diesel prices rises from 0.73 before the May 2021 transition to approximately 1.01 after it, and the increase is precisely estimated. The gasoline short-run interaction is small and statistically indistinguishable from no change.

That is not the whole picture for gasoline. Table 6 reports the error-correction model the cointegration diagnostic indicates. The long-run relationship is close to unit elastic ($\log \mathrm{PT} = 0.147 + 0.974 \log \mathrm{ES}$), and the speed at which Portuguese gasoline prices close a gap against that relationship roughly triples after the transition, from −0.091 to −0.299 per week. The implied half-life of a deviation falls from about eight weeks to a little over two. Gasoline linkage to Spain therefore did tighten; it tightened in the long-run adjustment channel rather than in contemporaneous pass-through, which a difference-only model cannot observe by construction.

Table 6: Gasoline error-correction model, two-step Engle–Granger, HAC(8), n = 1,077 weeks. Long-run relation $\log \mathrm{PT} = 0.147 + 0.974 \log \mathrm{ES}$. Source: `price_ecm_models.csv`.

Term	Estimate	Std. error	p
Disequilibrium (lagged)	−0.0907	0.0354	0.010
$\times$ post	−0.2080	0.0720	0.004
$\Delta \log$ ES	0.7889	0.0720	< 0.001
$\times$ post	0.1970	0.1648	0.232

The disequilibrium term is a generated regressor, so the second-stage standard errors are conditional on the first-stage cointegrating estimate; HAC covariance mitigates but does not remove this.

6 Portugal–Spain Comparison

The mean Portugal-minus-Spain pre-tax diesel spread moves from +4.12 to −26.66 EUR per 1000 litres across the transition, a change of −30.78; the gasoline spread moves by −18.80. These are descriptive. The diesel spread is not stationary (ADF $p = 0.605$), and the gasoline spread does not reject a unit root at five per cent either ($p = 0.066$), so neither spread can be treated as mean-reverting around a stable equilibrium, and spread levels should not be read as a disequilibrium measure.

Spain is used here as a comparison series, not as a causal control. No pre-trend or common-shock diagnostics have been estimated that would license a difference-in-differences interpretation, and both countries were exposed to the same European energy shock in 2022.

7 Source Reconciliation

JODI and DGEG are the only genuinely independent pair. Across the 24 overlapping 2019–2024 trade cells, 66.7% agree outright within tolerance. Eight cells diverge and each is recorded with the series that Eurostat identifies as the outlier: six where JODI is out of line and two where the DGEG workbook is. Over the full window, JODI and Eurostat agree on 89% of 80 cells, with only three of the nine divergences occurring before 2019.

A row is treated as divergent only when it breaches both the absolute and the percentage tolerance. Applying them as alternatives makes a 25 kt floor bind on series of roughly 1,600 kt, an effective tolerance of 1.6%, and flags agreement as close as 2.1% as failure.

8 Limitations

1. **2022 confounds the post-closure window.** The Matosinhos transition and the European energy shock are fourteen months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance and the Russian VGO disruption are competing explanations that are not separately identified.
2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result in this report is an association measured around a documented event.
3. **DGEG trade covers only 2019–2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission rather than being independent of it.
4. **Low annual power after 2021.** Three post-event annual observations cannot detect anything but a very large break, so the null 2022 annual results are uninformative rather than negative.
5. **Gasoline prices need an ECM.** The cointegration diagnostic indicates an error-correction specification that has not been estimated, so the reported gasoline short-run coefficients are incomplete.
6. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose semantics have not been confirmed against JODI documentation.
7. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. Pass-through results describe retail co-movement, not refinery-gate or international wholesale transmission.

9 Conclusions

The 2013 Sines hydrocracker produced an unambiguous structural change. Diesel refinery output and exports broke upward at 2013 and Portugal moved from net diesel importer to net diesel exporter, a result that survives multiplicity adjustment across sixteen tests.

After May 2021 the direction reverses, and by 2023 domestic diesel manufacturing covers only two thirds of demand, the lowest coverage in twenty years. A monthly design separates the closure from the 2022 energy shock and finds both to be independently associated with lower refinery output and higher net import dependence, the 2022 effect being the larger.

Short-run pass-through from Spanish to Portuguese pre-tax diesel prices rises from 0.73 to approximately 1.01 across the transition. Gasoline contemporaneous pass-through is unchanged, but its error-correction speed roughly triples, so both fuels became more tightly linked to Spanish prices through different channels. The initial reading, that only diesel changed, was an artefact of applying a difference-only model to a cointegrated pair: that specification cannot see the long-run adjustment channel where the gasoline change actually appears.

This is an association, not a demonstrated causal effect. The closure and the largest European energy shock in decades fall within fourteen months of one another, and this design cannot separate their contributions to the post-2021 level. What the evidence supports is that Portugal's diesel system moved from surplus to deficit over the study window, that the movement is concentrated around two documented events, and that Portuguese diesel prices now track Spanish diesel prices considerably more closely than before.

Claim–Evidence Matrix

Claim	Evidence file	Evidence level	Caveat
Sines hydrocracker entered commercial production in 2013	`refinery_events.csv`	documented event	Corporate disclosure, not outcome data
Matosinhos refining ceased May 2021	`refinery_events.csv`	documented event	2021 treated as a transition year
Diesel net-import ratio breaks at 2013	`structural_break_tests.csv`	structural break	Diagnostic, not a causal estimator
Portugal was a net diesel exporter 2013–2020	`fuel_annual_analytical_panel.csv`	descriptive	Ratio is not self-sufficiency
Refinery output falls 105 kt/month at the closure boundary	`monthly_event_models.csv`	association	Quote with its phase-trend term
2022 stress adds a further −152 kt/month	`monthly_event_models.csv`	association	Confounded with demand recovery
2022 diesel exports at the 0th baseline percentile	`stress_2022_metrics.csv`	descriptive	Baseline window chosen ex ante
Diesel pass-through rises 0.73 to 1.01	`price_short_run_models.csv`	association	Retail, not wholesale, prices
Gasoline pass-through unchanged	`price_short_run_models.csv`	association	ECM indicated but not estimated
PT–ES diesel spread moves −30.8 EUR/1000L	`pt_es_price_spread_summary.csv`	descriptive	Spread is non-stationary
Sources agree on 89% of full-window trade cells	`jodi_dgeg_trade_reconciliation.csv`	descriptive	Eurostat is DGEG-derived
No causal effect of the closure is claimed	—	—	2022 is not separable by design